TeachBack

Safety-weighted, component-level scoring of patients' medication teach-back

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LLM Applications course ▪ Bar-Ilan University ▪ Project proposal

Slide 1: Motivating use case

Use case background

- *Teach-back*: the patient repeats the instructions in their own words
- Nurses and pharmacists, mostly at discharge

Why it is important

- Misunderstanding → harm at home
- Wrong dose ▪ stopping early ▪ alcohol

Why it is challenging

- Free, messy language with many paraphrases
- Not all mistakes are equal
- Only 1 in 5 patients right on the first try
- No labelled data

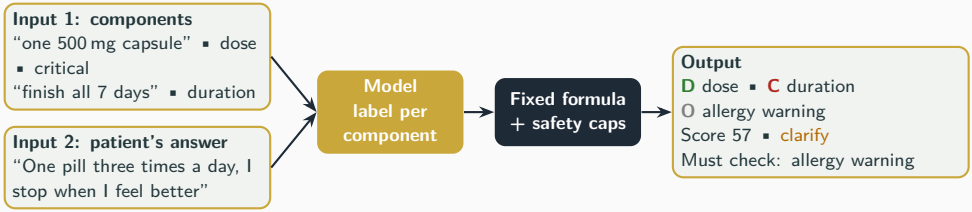
How it is solved today

- The clinician judges by ear; nothing is recorded
- Research: questionnaires and multiple choice

Slide 2: Project task description

Formal problem statement

For each instruction component, decide if the patient's answer shows it as **D**emonstrated, **P**artial, **O**mitted or **C**ontradicted; then compute a safety-weighted score.



Scoring rule

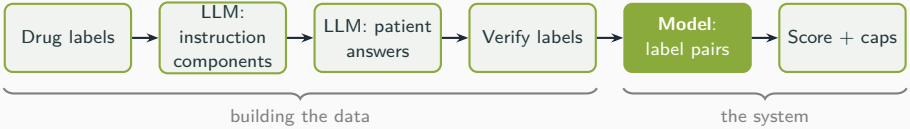
- **D** 1 ▪ **P** 0.5 ▪ **O** 0 ▪ **C** $-0.5 \times$ harm weight
- Critical: **C** $\Rightarrow \leq 40$ ▪ **O**/**P** $\Rightarrow \leq 84$
- $\geq$85 understood ▪ 50–84 clarify ▪ $<$50 re-teach

Project novelty

- Scores the patient's *own words*
- Per component, weighted by harm
- A dangerous error can't hide in a high score

Slide 3: Models and methods

Processing pipeline



Off-the-shelf

- **E0** Word overlap
- **E1** LLM judge
- **E2** Zero-shot NLI

Fine-tuned

- **E3a** DeBERTa-v3 *(main)*
- **E3b** ModernBERT
- **E3c** Score regression

Adjustments

- Start from NLI checkpoint
- Class-weighted loss
- Early stopping, 3 seeds
- E1: rubric + few-shot

Slide 4: Data specification and generation

Data requirements

- A known label for every component
- Many writing styles
- Tricky errors and valid paraphrases

Labeling

- Label chosen at generation time
- Checked by a second LLM + by hand
- 150 role-played answers, 2 annotators

Dataset

- Synthetic, from DailyMed / openFDA labels
- $\approx$3,200 answers $\rightarrow$ $\approx$16–19k pairs
- Split by instruction; unseen drug classes

Synthetic generation

- Ask the LLM for a label per component + a style
- Inject errors: 50 mg for 500, “a little alcohol is fine”

Slide 5: Metrics and KPIs

Main KPIs

- **Macro-F1**
  per component
- **Dangerous misses**
  target ≈ 0
- **Omission misses**
  target ≈ 0

Quality of each step

- Data: generator fidelity
- Model: F1 per label
- Score: Spearman, MAE, QWK
- Safety: over-flag rate
- Robustness: same meaning $\rightarrow$ same score

Protocol and ground truth

- Verified labels
- Human set, κ agreement
- 3 test sets
- 3 seeds, mean $\pm$ std
- Weight sensitivity

Backup

Backup: worked example

Instruction: "Take one 500 mg amoxicillin capsule three times a day for 7 days. Finish all of it even if you feel better. Call your doctor right away if you get a rash, swelling or trouble breathing."

Restatement: "I take one capsule three times a day, and I can stop once I feel better."

Component	Category	Critical	$\tilde{w}_i$	Status	Points
One 500 mg capsule per dose	dose	yes	0.357	**D** ("one capsule")	+0.357
Three times a day	timing	no	0.286	**D**	+0.286
For 7 days	duration	no	0.071	**C** ("stop once I feel better")	−0.036
Finish even if better	duration	no	0.071	**C**	−0.036
Rash / swelling / breathing → call doctor	warning	yes	0.214	**O**	0

$S = 57 \Rightarrow$ **clarification recommended**. **Must check**: allergy warning signs (critical, omitted; the 84 cap holds but 57 is already lower). Revisit: course length.

If "finish the course" were flagged critical: $\min(57, 40) = 40 \Rightarrow$ substantial misunderstanding. Dose wrong but all else right, in a 7-category instruction: 62.5 without the cap $\Rightarrow$ capped to 40. Only "no alcohol" (weight 0.10, critical) omitted, all else right: 90 $\Rightarrow$ capped to 84, clarification recommended.

Backup: scoring rule details and scope

Category base weights

dose	0.25	warning signs	0.15
timing	0.20	duration	0.10
interactions	0.20	administration	0.05
		missed dose	0.05

Split equally among a category's components, then renormalised over the categories present. Credit: $D = 1$, $P = 0.5$, $O = 0$, $C = -0.5$ (a wrong belief is worse than none).

Safety caps: critical **C** $\Rightarrow S \leq 40$; critical **O**/**P** $\Rightarrow S \leq 84$. "No flagged" requires every critical component to be **D**. Omission = missing evidence, not proven error, hence the softer cap.

Weights, credits, caps and thresholds are *hypotheses*, reviewed by a pharmacist or nurse if available, otherwise based on label text and the ISMP high-alert list.

Scope

- Measures *demonstrated* understanding in the restatement, not what the patient knows.
- Out of scope: splitting free-text instructions, speech, corrective advice.
- Clinician in the loop; synthetic data only.

Closest work

- Teach-back studies: recall scored by hand.
- DischargeSim (2025): multiple-choice comprehension exams.
- HealthLit (CHASE 2025): scores the clinician's practice.
- Short-answer grading: no safety weighting.